

Problem Solving

Unit I Introduction to problem solving, and set theory

Lecture 7

Introduction to algorithmic problem solving, review of set theory, invariants, chocolate bars, empty boxes, the tumbler problem, crossing a river: brute force, nervous couples, the bridge problem and conditional statements

Unit II Games

Lecture 7

Games: matchstick games, winning strategies: assumptions, labelling positions, winning strategies: formulating requirements, subtraction-set games, sums of games: a simple sum game, sums of games: maintain symmetry, more simple sums, sums of games: evaluating positions, using the Mex function.

Unit III Knights and Knaves

Lecture 7

Knights and knaves: logic puzzles, calculation logic: propositions, knight and knaves, calculation logics: Boolean equality, hidden treasures, calculation logics, equals for equals, equivalence and continues equalities, negation: contraposition, negation: handshake problem, inequivalence.

Unit IV Induction, Fake coin detection and Tower of Hanoi

Lecture 9

Induction: example problems, cutting the plane, induction: Triominoes, looking for patterns induction: the need for proof, from verification to construction, fake-coin detection: problem formulation, fake-coin detection: problem solution, the tower of Hanoi: specification and solution, the tower of Hanoi: inductive solution, the iterative solution, the language of mathematics: variables, expressions and laws, the language of mathematics: functions, algebraic properties, Boolean operators, and binary relations.

Textbook:

1. Roland Bachkhouse, "Algorithmic Problem Solving", Publisher Wiley, 2011, ISBN:9780470684535.

Session Plan

Lecture	Topic	Remarks
1	Introduction to problem solving and review of set theory	
2	Invariants, Chocolate Bars	
3	Empty boxes	
4	The Tumbler Problem	
5	Crossing a River: Brute Force	
6	Nervous Couples	
7	The Bridge Problem and Conditional Statements	
8	Games: Matchstick games	
9	Winning Strategies: Assumptions, Labelling Positions	
10	Winning Strategies: Formulating Requirements	
11	Subtraction-Set Games	
12	Sums of Games: A simple sum game	
13	Sums of Games: Maintain Symmetry, more simple sums	
14	Sums of Games: Evaluating positions, Using the Mex function	
15	Knights and Knaves: Logic puzzles	
16	Calculation Logic: Propositions, Knight and Knaves	
17	Calculation Logics: Boolean Equality, Hidden Treasures	
18	Calculation Logics: Equals for Equals	
19	Equivalence and Continues Equalities	
20	Negation: Contraposition	
21	Negation: Handshake Problem, Inequivalence	
22	Induction: Example Problems, Cutting the Plane	
23	Induction: Triominoes, Looking for Patterns	
24	Induction: The Need for Proof, From Verification to Construction	
25	Fake-Coin Detection: Problem Formulation	
26	Fake-Coin Detection: Problem Solution	
27	The Tower of Hanoi: Specification and Solution	
28	The Tower of Hanoi: Inductive Solution, The Iterative Solution	
29	The Language of Mathematics: Variables, Expressions and Laws	
30	The Language of Mathematics: Functions, Algebraic Properties, Boolean Operators, and Binary Relations	